

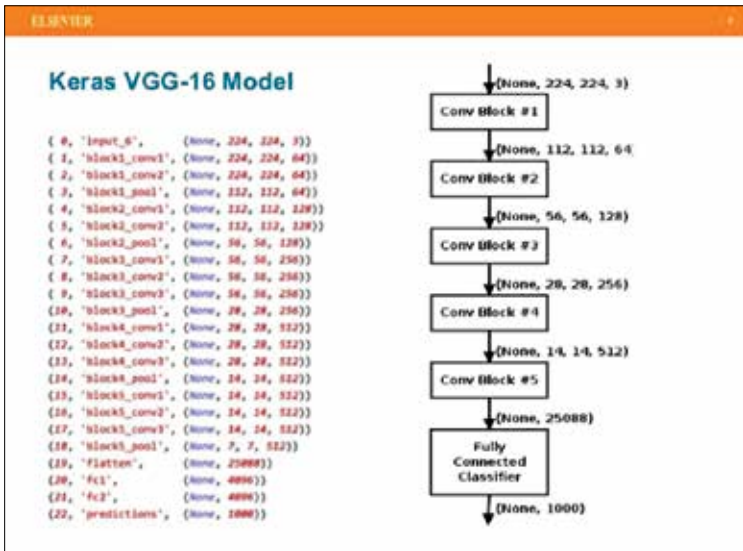


Figure : VGG 16 Model Architecture

we created. Since we want a convolution layer, we pass a Conv2D layer to the input. This first layer is called the input layer. It has some important parameters we need to set like:

- 1.1. Filter size: This is the number of output filters in the convolution layer. It is the size of the output dimension. We will be using the afilter size of 32 in this layer.
- 1.2. kernel_size: It specifies the height and width of the 2D convolution window. We would be using a height and width size of 3,3.
- 1.3. Activation function: An activation function is also called non - linearity because they introduce non-linear properties to our Neural Network. Their main purpose is to convert an input signal of a node in an artificial neural network to an output signal. That output signal is now used as an input for the next layer in the stack. Rectified Linear Unit or ReLU is the most common activation function used. Other variations of ReLU are leaky ReLU and eLU. However, in our model, we are going to use the ReLU activation function.
- 1.4. Input shape: We had resized our images to 150 by 150 (height by width) including a channel of three. Hence, we would be using the input shape to be [150, 150, 3] that specifies the height, width and the number of channels